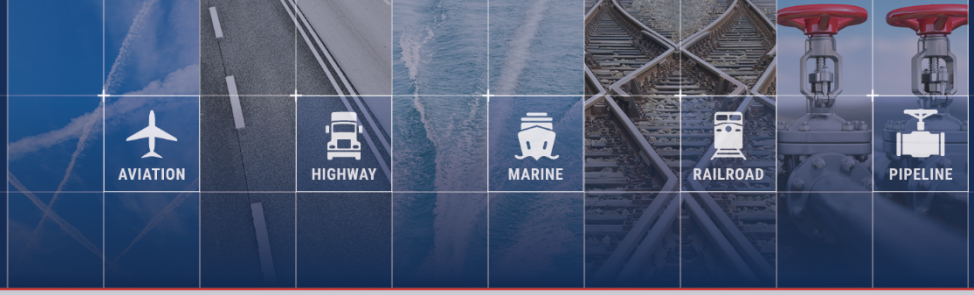

Aviation Investigation Final Report

Location:	Gallup, New Mexico	**Accident Number:**	WPR25LA071
Date & Time:	December 27, 2024, 21:03 Local	**Registration:**	N505SW
Aircraft:	Cessna T210L	**Aircraft Damage:**	Substantial
Defining Event:	Fuel related	**Injuries:**	2 None
Flight Conducted Under:	Part 91: General aviation - Personal		

Analysis

Before departing on a cross-country flight, the pilot fueled the airplane with about 20 gallons of fuel, using the airplane's gauges to determine the airplane's fuel quantity, which she believed to be 70 gallons. The pilot did not look inside the fuel tanks to verify the fuel quantity before departure. After about two hours of flight, while on approach to the destination airport, the engine lost total power and the pilot was unable to restore engine power or glide to the runway. During the subsequent forced landing in a vacant lot, the airplane impacted powerlines and rough terrain, resulting in substantial damage to the fuselage and right wing.

According to the pilot, an unknown quantity of fuel was observed to leak from the airplane's lower forward fuselage after the accident. During the airplane's recovery from the accident site, only trace amounts of fuel were found within the fuel system. Postaccident examination of the airplane revealed no evidence of a preexisting malfunction or failure that would have precluded normal operation, and a test run of the engine revealed no anomalies. The left-side fuel system was unobstructed and undamaged with exception of the left-side fuel reservoir tank, located in the lower forward fuselage, which displayed fuel staining and damage consistent with impact. No damage or voids were found in the right-side fuel system.

As there were no mechanical anomalies with the engine or right-side fuel system, and there was no evidence to suggest that the left-side fuel system was compromised before impact, the engine likely lost power due to the pilot's improper fuel management. The penetrated left side fuel reservoir tank was located at a low point in the fuel system relative to the left-wing fuel tank and leaked an unknown quantity of fuel at the accident site. The investigation could not determine if the loss of power was due to fuel starvation or exhaustion, as the remaining fuel quantity in the left tank at the time of the accident is unknown.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be:

The pilot's improper fuel management, which resulted in a total loss of engine power.

Findings

Aircraft	Fuel - Fluid level
Personnel issues	Preflight inspection - Pilot
Personnel issues	Decision making/judgment - Pilot
Personnel issues	Fuel planning - Pilot
Environmental issues	Wire - Contributed to outcome

Factual Information

History of Flight

Approach-VFR pattern final	Fuel related (Defining event)

On December 27, 2024, at 2103 mountain standard time, a Cessna T210L, N505SW, was substantially damaged when it was involved in an accident near Gallup Municipal Airport (GUP), Gallup, New Mexico. The pilot and one passenger were not injured. The airplane was operated as a Title 14 *Code of Federal Regulations* Part 91 personal flight.

According to the pilot, earlier on the day of the accident, she purchased 56.6 gallons of fuel for the airplane. The pilot stated that the fuel gauges indicated both fuel tanks were full, but she did not visually verify the fuel level in the tanks. The pilot and passenger departed GUP and flew to Lake Havasu City Airport (HII), Lake Havasu City, Arizona, with an enroute stop at Cottonwood Airport (P52), Cottonwood, Arizona. The pilot stated these two flight legs totaled 3 flight hours as indicated by the airplane's tachometer.

While at HII, the pilot used the self-serve fuel pump to add about 10 gallons of 100LL fuel to each of the airplane's two wing fuel tanks. The pilot used the airplane's fuel gauges to determine the fuel quantity, stating that each gauge indicated 35 gallons of fuel, or 70 total gallons. The pilot did not visually verify the fuel level in the tanks. The pilot and passenger subsequently departed on the return flight to GUP about 1900.

According to the pilot, the flight from HII to GUP took two hours. As the airplane entered the base turn to land on runway 24 at GUP, the pilot noted that the airplane was descending faster than expected and added engine power; however, the engine did not respond. The pilot verified the position of the engine controls and fuel selector and noted each fuel gauge appeared to be half-full. The pilot stated the airplane was unable to reach the runway and initiated a forced landing in a vacant lot about 1,800 ft west of runway 24. During the forced landing, the airplane impacted power lines and rough terrain, which resulted in substantial damage to the right wing and fuselage.

Two days after the accident, the airplane was moved from the vacant lot to a parking lot at GUP. The pilot was present when the airplane was moved and stated that an unknown quantity of fuel leaked from the airplane's belly. According to the airport manager, only trace amounts of fuel were observed within the airplane's fuel system.

About two months after the accident, the airplane was transported from GUP to a storage facility. Personnel who disassembled the airplane drained fuel from each of the wing tanks; they estimated about 3 to 4 ounces of fuel was recovered from the left wing fuel tank, and

about 16 ounces of fuel was recovered from the right wing fuel tank. The fuel selector was found set to the left fuel tank, and impact damage prevented it from being turned by hand.

Examination of the airplane and engine revealed no evidence of preimpact malfunctions or anomalies. The left fuel collector tank was impact-damaged and displayed a small crack on the upper surface, less than .125 inch long, that was ringed with blue fuel stains. No other leaks or breaches were found in the airplane's fuel system. The engine was subsequently test run, during which it operated smoothly and continuously through various power settings.

The wings were each raised using a forklift and moved adjacent to their respective wing roots. The wiring for the fuel quantity indicators was reconnected, and the wings were situated to simulate dihedral. Fuel was added to each wing fuel tank in increments, and the gauge was read at each increment. The following observations were made:

Fuel Added to Each Tank	Reading on Left Gauge	Reading on Right Gauge
Empty	0	0
5 gallons	4 gallons	5 gallons
15 gallons	9 gallons	14 gallons
20 gallons	17 gallons	20 gallons

The airplane's fuel storage system consisted of two vented integral fuel tanks (one in each wing) and two fuel reservoir (collector) tanks. Fuel flowed by gravity from the two integral tanks to two reservoir tanks, and from the reservoir tanks to a three-position selector valve labeled LEFT ON, RIGHT ON, and OFF. Fuel cannot be used from both tanks simultaneously. Of the airplane's 90-gallon fuel capacity, one gallon (or .5 gallon per tank) is considered unusable, and cannot be safely used in flight. Additionally, the fuel consumption rate ranges from 6 to 20 gallons per hour in normal operating conditions, with 20 to 28 gallons per hour during takeoff and climb.

Pilot Information

Certificate:	Commercial; Flight instructor	**Age:**	26,Female
Airplane Rating(s):	Single-engine land; Multi-engine land; Multi-engine sea	**Seat Occupied:**	Left
Other Aircraft Rating(s):	Unmanned (sUAS)	**Restraint Used:**	3-point
Instrument Rating(s):	Airplane	**Second Pilot Present:**	No
Instructor Rating(s):	Airplane single-engine; Instrument airplane	**Toxicology Performed:**	
Medical Certification:	Class 1 Without waivers/limitations	**Last FAA Medical Exam:**	November 1, 2023
Occupational Pilot:	Yes	**Last Flight Review or Equivalent:**	November 1, 2023
Flight Time:	(Estimated) 981.4 hours (Total, all aircraft), 49.4 hours (Total, this make and model), 812.9 hours (Pilot In Command, all aircraft), 96.7 hours (Last 90 days, all aircraft), 23.2 hours (Last 30 days, all aircraft), 5 hours (Last 24 hours, all aircraft)		

Passenger Information

Certificate:		**Age:**	
Airplane Rating(s):		**Seat Occupied:**	Right
Other Aircraft Rating(s):		**Restraint Used:**	3-point
Instrument Rating(s):		**Second Pilot Present:**	No
Instructor Rating(s):		**Toxicology Performed:**	
Medical Certification:		**Last FAA Medical Exam:**	
Occupational Pilot:	No	**Last Flight Review or Equivalent:**	
Flight Time:			

Aircraft and Owner/Operator Information

Aircraft Make:	Cessna	**Registration:**	N505SW
Model/Series:	T210L	**Aircraft Category:**	Airplane
Year of Manufacture:	1976	**Amateur Built:**	
Airworthiness Certificate:	Normal	**Serial Number:**	21061185
Landing Gear Type:	Retractable - Tricycle	**Seats:**	6
Date/Type of Last Inspection:	Annual	**Certified Max Gross Wt.:**	3800 lbs
Time Since Last Inspection:		**Engines:**	1 Reciprocating
Airframe Total Time:		**Engine Manufacturer:**	CONT MOTOR
ELT:	C126 installed, activated, did not aid in locating accident	**Engine Model/Series:**	TSIO-520-H
Registered Owner:	On file	**Rated Power:**	285 Horsepower
Operator:	On file	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual (VMC)	**Condition of Light:**	Night/dark
Observation Facility, Elevation:	KGUP,6464 ft msl	**Distance from Accident Site:**	1 Nautical Miles
Observation Time:	21:53 Local	**Direction from Accident Site:**	258°
Lowest Cloud Condition:	Clear	**Visibility**	10 miles
Lowest Ceiling:	None	**Visibility (RVR):**	
Wind Speed/Gusts:	8 knots / None	**Turbulence Type Forecast/Actual:**	None / None
Wind Direction:	270°	**Turbulence Severity Forecast/Actual:**	N/A / N/A
Altimeter Setting:	30 inches Hg	**Temperature/Dew Point:**	4°C / -5°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Lake Havasu City, AZ (HII)	**Type of Flight Plan Filed:**	None
Destination:	Gallup, NM	**Type of Clearance:**	None
Departure Time:	19:10 Local	**Type of Airspace:**	Class G

Airport Information

Airport:	Gallup Municipal Airport GUP	**Runway Surface Type:**	Asphalt
Airport Elevation:	6472 ft msl	**Runway Surface Condition:**	Dry;Rough
Runway Used:	24	**IFR Approach:**	None
Runway Length/Width:	7312 ft / 100 ft	**VFR Approach/Landing:**	Traffic pattern

Wreckage and Impact Information

Crew Injuries:	1 None	**Aircraft Damage:**	Substantial
Passenger Injuries:	1 None	**Aircraft Fire:**	None
Ground Injuries:		**Aircraft Explosion:**	None
Total Injuries:	2 None	**Latitude, Longitude:**	35.512773,-108.772

Administrative Information

Investigator In Charge (IIC):	Whitaker, Kathryn
Additional Participating Persons:	Douglas Peterson; Federal Aviation Administration; Albuquerque, NM Casey Love; Textron Aviation; Wichita, KS
Original Publish Date:	November 5, 2025
Last Revision Date:	
Investigation Class:	Class 3
Note:	The NTSB did not travel to the scene of this accident.
Investigation Docket:	https://data.ntsb.gov/Docket?ProjectID=199497

The National Transportation Safety Board (NTSB) is an independent federal agency charged by Congress with investigating every civil aviation accident in the United States and significant events in other modes of transportation—railroad, transit, highway, marine, pipeline, and commercial space. We determine the probable causes of the accidents and events we investigate, and issue safety recommendations aimed at preventing future occurrences. In addition, we conduct transportation safety research studies and offer information and other assistance to family members and survivors for each accident or event we investigate. We also serve as the appellate authority for enforcement actions involving aviation and mariner certificates issued by the Federal Aviation Administration (FAA) and US Coast Guard, and we adjudicate appeals of civil penalty actions taken by the FAA.

The NTSB does not assign fault or blame for an accident or incident; rather, as specified by NTSB regulation, "accident/incident investigations are fact-finding proceedings with no formal issues and no adverse parties … and are not conducted for the purpose of determining the rights or liabilities of any person" (Title 49 *Code of Federal Regulations* section 831.4). Assignment of fault or legal liability is not relevant to the NTSB's statutory mission to improve transportation safety by investigating accidents and incidents and issuing safety recommendations. In addition, statutory language prohibits the admission into evidence or use of any part of an NTSB report related to an accident in a civil action for damages resulting from a matter mentioned in the report (Title 49 *United States Code* section 1154(b)). A factual report that may be admissible under 49 *United States Code* section 1154(b) is available here.